Theoretical and Applied Mechanics Letters 16 (2026) 100658

Contents lists available at ScienceDirect

Theoretical and Applied Mechanics Letters

journal homepage: www.elsevier.com/locate/taml

Research Article

Development of a large language model-driven intelligent agent for predicting relative permeability in oil and gas reservoirs

Jiulong Wang [a,b], Yuanchun Zhou [b,c], Xuhui Zhang [d], Junming Lao [a], Hongqing Song [a,b,*]

[a] School of Resource and Safety Engineering, University of Science and Technology Beijing, Beijing 100083, China
[b] National & Local Joint Engineering Laboratory for Big-data Analysis and Computing Technology, Beijing 100083, China
[c] Computer Network Information Center, Chinese Academy of Sciences, Beijing 100083, China
[d] Institute of Mechanics, Chinese Academy of Sciences, Beijing 100190, China

ARTICLE INFO

Keywords:
Relative permeability
Large language model
Intelligent agent

ABSTRACT

To address the challenges and difficulties in predicting the relative permeability of reservoirs using traditional physics-driven and data-driven approaches, this paper proposes a collaborative analysis intelligent agent for the relative permeability of oil and gas reservoirs based on a large model. By constructing a multiagent collaborative workflow, integrated collaboration of data, models, and analysis results is achieved. The intelligent agent automatically completes data preprocessing, feature extraction, parameter calibration, small model calling, and output and evaluation of prediction results based on preset task dependencies. At the same time, by introducing deep learning-based embedding of physical information, the analysis efficiency and accuracy are significantly improved. The results show that compared with traditional physical analysis methods, this method improves the accuracy of reservoir relative permeability prediction by 10%, has a computational efficiency 10 times higher than traditional deep learning algorithms, and a computational speed 100–1,000 times higher than conventional physical models. This study further enhances the efficiency and intelligence of physical property analysis of oil and gas reservoirs, providing a new research direction for the intelligent development of oil and gas digitization.

1. Introduction

The relative permeability prediction for oil and gas reservoirs is a cornerstone of hydrocarbon exploration and production, directly influencing reserve estimation, development-plan optimization, and recovery enhancement. Traditional prediction approaches have relied primarily on physical-model experiments, theoretical models based on capillary pressure and numerical simulation analysis [1,2]. These three methods have undergone long-term verification, possess clear physical mechanisms and are mature techniques. They exhibit high accuracy in the analysis of relatively homogeneous reservoirs and have played a significant role in reservoir flow analysis over the past few decades. However, with the dominance of unconventional reservoir development characterized by strong heterogeneity, traditional relative permeability analysis methods are facing the current situation of increased experimental costs and cycles, significantly increased analytical errors, low computational efficiency, and insufficient theoretical support.

In recent years, with the rapid development of artificial intelligence technology, machine learning methods such as deep learning have been utilized to mine and analyze multimodal data, including core images, core displacement experiments, and numerical simulations, to construct a data-driven reservoir relative permeability analysis model [3–13]. This method breaks through the theoretical iterative analysis and complex processes of traditional physical methods, enabling rapid calculation of reservoir relative permeability [14–21]. Despite these advances, the techniques still face five major bottlenecks:

(1) Insufficient multisource data integration. No effective mechanism yet unites well-log, seismic, geomechanical, and dynamic production data, which usually remain isolated from prediction models, leading to a lack of physical consistency in the outputs.
(2) Imbalance between long-sequence dependency and local feature capture. Recurrent neural networks and their variants still suffer from gradient decay when processing ultralong well-log sequences, while conventional convolutional neural networks (CNNs) lack temporal constraints and risk future-data leakage [22].
(3) Limited model generalization. The geological structures of China's basins are highly complex and laterally irregular—examples include the carbonate fracture systems of the Sichuan Basin and the low-permeability reservoirs of the Ordos Basin—so AI models trained on one region may generalize poorly to another. Moreover, the scarcity of high-quality labeled samples hampers precise prediction, yielding realized efficiency improvements of only 60%–70% of targets.

* Corresponding author at: School of Resource and Safety Engineering, University of Science and Technology Beijing, Beijing 100083, China.
E-mail address: songhongqing@ustb.edu.cn (H. Song).

https://doi.org/10.1016/j.taml.2026.100658
Received 25 July 2025; Received in revised form 20 January 2026; Accepted 22 January 2026
Available online 23 January 2026

(4) Computational efficiency and real-time constraints. Drilling to deep reservoirs (depths > 4,500 m) increases well costs by over 380%, creating an urgent need for rapid prediction methods. Traditional workflows struggle to process terabytes of seismic data efficiently, limiting real-time decision support.
(5) "Black-box" risks due to missing physical mechanisms. Purely data-driven models lack embedded geomechanical understanding, resulting in poor interpretability and low trust among reservoir engineers. The current industry trend has shifted toward a hybrid paradigm of "physical mechanism + data-driven", which retains physical interpretability while breaking through the experimental and computational bottlenecks of traditional methods through AI technology [23].

However, the rapid development of large language models (LLMs) has provided a new direction for solving the abovementioned challenges. In 2020, OpenAI released GPT-3, which was a groundbreaking event that marked the true emergence of LLMs in the field of natural language processing, officially ushering in the era of LLMs. As a type of deep learning artificial intelligence model, LLMs are trained on large-scale datasets to generate natural language text or understand the meaning of language text. The LLMs adopt a transformer architecture and pretraining objectives similar to the small model, with the main difference being the increase in model size, training data, and computing resources. Compared to traditional natural language processing (NLP) models, LLMs can better understand and generate natural text while exhibiting stronger logical thinking and reasoning abilities. In recent years, LLMs have also shown significant advantages in the oil and gas development industry, especially in processing complex heterogeneous reservoirs, multisource data fusion, and improving prediction accuracy, for example, using LLMs to reconstruct or fill missing logging curves with high accuracy based on neighboring curves or seismic data. Retrieve elastic parameters (such as longitudinal wave velocity, transverse wave velocity, and density) and rock physical parameters (such as porosity and mud content) from seismic data. By integrating multiple parameters, such as physical properties, oil and gas content, and fracturability, LLMs are used to identify "sweet spots" where oil and gas are enriched. Combined with sedimentary facies information, the distribution of physical property parameters in different facies zones is predicted [24–31]. Currently, the most successful research and application of LLMS are in the United States and China. The latest global rankings of LLMs are shown in Table 1.

The layout of global oil companies in the field of large models is accelerating, with China Petroleum, Sinopec, and CNOOC becoming industry leaders. International oil giants such as Shell BP and ExxonMobil are also exploring cooperation and application promotion with tech giants such as Google and Microsoft. The China National Petroleum Corporation released the Kunlun Model with 33 billion parameters in 2024 and upgraded it to 300 billion parameters in 2025. Its application scenarios include seismic interpretation, well logging processing, etc. Sinopec's Shengxiaoli Da model has 93 billion parameters and has reinforced the learning of 600,000 pieces of oil and gas professional knowledge. The ocean energy model of the China National Offshore Oil Corporation covers over 100 business scenarios.

To address the challenges and difficulties in predicting the relative permeability of reservoirs using traditional physics-driven and data-driven approaches, this paper proposes a collaborative analysis intelligent agent for the relative permeability of oil and gas reservoirs based on a large model. By constructing a multiagent collaborative workflow, integrated collaboration of data, models, and analysis results is achieved. The intelligent agent automatically completes data preprocessing, feature extraction, parameter calibration, small model calling, and output and evaluation of prediction results based on preset task dependencies. At the same time, by introducing deep learning-based embedding of physical information, the analysis efficiency and accuracy are significantly improved. The results show that compared with traditional physical analysis methods, this method improves the accuracy of reservoir relative permeability prediction by 10%, has a computational efficiency 10 times higher than traditional deep learning algorithms, and a computational speed 100–1,000 times higher than conventional physical models.

2. Intelligent analysis of relative permeability considering physical properties

To enable intelligent agents to gain a deeper understanding of the pore structure of reservoirs and the impact mechanism of microscopic effects on relative permeability, this study simulates the flow process of a 3-digit digital core through a pore network model (PNM), analyzes and obtains the relative permeability of the core, and then uses artificial intelligence algorithms to perform multimodal training on the digital core images and analysis results. Finally, an intelligent agent for predicting relative permeability is constructed through the generalization ability and semantic reasoning ability of the large model.

2.1. The improved pore network model

In this paper, we use an improved PNM to obtain relative permeabilities as output data for our AI training and testing processes. The improved model successfully considered the viscous coupling effect by integrating the direct single-pore simulation results from the lattice Boltzmann method (LBM); thus, it shows higher accuracy than the original network model [32,33].

In the PNM, the relative permeability K_r is calculated by the ratio of flow rates q_tm and q_ts, as shown in Eq. (2.1):

$$K_\mathrm{r} = \frac{q_\mathrm{tm}}{q_\mathrm{ts}}, \tag{2.1}$$

where q_tm and q_ts are the total flow rates in multiphase and single flow conditions, respectively. The fluids are considered to be incompressible, and the total flow rate is computed by solving for the pressure everywhere, applying mass balance at every pore i,

$$\sum_j q_{,ij} = 0, \tag{2.2}$$

Table 1
Global LLMS model ranking. (2025-06, https://arena.lmsys.org/).

Modle	Artificial analysis intelligence index	Context window (k)	MMLU-PRO (%)	Organization	License
Grok-4	73	256	87	xAI	Proprietary
o3-pro	71	200		OpenAI	Proprietary
Gemini-2.5-Pro	70	1	86	Google	Proprietary
o3	70	200	85	OpenAI	Proprietary
o4-mini (high)	69	200	83	OpenAI	Proprietary
Gemini-2.5-Pro (Mar'25)	68	1	86	Google	Proprietary
DeepSeek R1 0528 (Mar'25)	68	128	85	DeepSeek	Open
Gemini-2.5-Pro (May'25)	68	1	84	Google	Proprietary
Grok 3 mini Reasoning (high)	67	1	83	xAI	Proprietary
o3-mini (high)	66	200	80	OpenAI	Proprietary
Gemini 2.5 Flash (Reasoning)	65	1	83	Google	Proprietary
Claude 4 Opus Thinking	64	200	87	Anthropic	Open

where j represents each throat connected to pore i. By assuming that throat j connects pores i and k, the flow rate q_{ik} between two adjacent pores i and k is given by:

$$q_{ik} = g_{ik} \nabla P_{i,k}, \tag{2.3}$$

where $\nabla P_{i,k}$ and g_{ik} represent the pressure gradient and flow conductance between pores i and k, respectively. If the flow is capillary pressure dominated, the pressure can be given as the capillary pressure P_c by the Young-Laplace equation:

$$P_\mathrm{c} = \gamma \kappa, \tag{2.4}$$

where γ is the interfacial tension and κ represents the interface curvature. The expressions of flow conductance for pore-throat units with different configurations are given as:

$$g_\mathrm{c} = C \frac{A_\mathrm{c}^2 G_\mathrm{c}}{\mu_\mathrm{c}} \cdot f_\mathrm{c}\left(\frac{\mu_\mathrm{corner}}{\mu_\mathrm{center}}\right), \tag{2.5}$$

$$g_\mathrm{l} = \frac{b_\mathrm{o}^4 \tilde{g}_\mathrm{l}}{\mu_\mathrm{l}} \cdot f_\mathrm{l}\left(\frac{\mu_\mathrm{layer}}{\mu_\mathrm{corner}}\right), \tag{2.6}$$

where the subscripts c and l represent the corner and layer configurations, respectively; μ means the dynamic viscosity; A_c denotes the corner area; G_c, C, b_o, and $\tilde{g}_\mathrm{l}$ are geometric parameters; $f_\mathrm{c}(\frac{\mu_\mathrm{corner}}{\mu_\mathrm{center}})$ and $f_\mathrm{l}(\frac{\mu_\mathrm{layer}}{\mu_\mathrm{corner}})$ are empirical functions to consider the viscous coupling effect for the corner and layer configurations, respectively.

Through the above model simulation analysis, we can obtain the relative permeability of the rock core, which provides a dataset for intelligent prediction models.

2.2. Intelligent prediction model for relative permeability

In Section 2.1, we use a digital core to calculate the relative permeability of the core through a pore network model. To obtain the relative permeability curve of the core more quickly and efficiently, this paper constructs a deep learning method to learn the correlation features between the 3D digital core structure characteristics, pore network model, and corresponding relative permeability results, thereby achieving rapid analysis from the core to the relative permeability.

To obtain spatial features in 3D images more accurately, this paper adopts the convolutional long short-term memory-convolutional neural network (ConvLSTM-CNN) method. On the basis of leveraging the advantages of CNNs in extracting spatial features, ConvLSTM introduces a time/sequence memory gating mechanism, which enables the model to learn the dynamic evolution and dependency relationship of feature values in the 3D spatial sequence direction (such as depth direction and time step of simulating fluid flow), ensuring that the integrity of spatial information is maintained during the spatiotemporal evolution modeling process. The model architecture diagram is shown in Fig. 1. At the same time, ConvLSTM can learn complex temporal dynamic patterns based on these spatial representations, predict future states (such as fluid distribution at the next moment), or understand the evolution laws of processes. The results show that the ConvLSTM-CNN method has significant advantages in feature extraction performance compared to the CNN method. Compared to physical simulations, this model can provide more efficient predictions, especially in situations where parameters are uncertain or boundary conditions are complex.

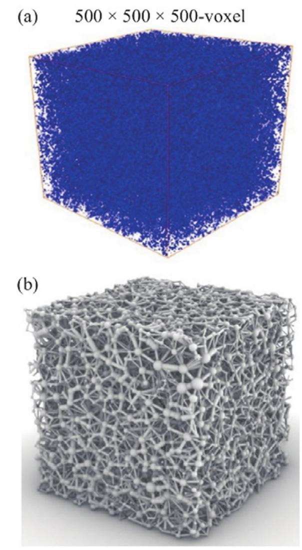

Fig. 2. (a) 3D digital rock cores. (b) Extracted pore networks.

In this article, we selected a total of 1000 3D digital rock images for analysis. To ensure consistency of analysis results and control simulation time, all of the 1000 images have the same voxel size of 500 × 500 × 500, as shown in Fig. 2(a). Then, based on 1000 3D digital core images, 1000 pore network models were constructed, as shown in Fig. 2(b), and the relative permeability values of 1000 groups were simulated and calculated.

Based on the ConvLSTM-CNN algorithm, 1000 sets of digital rock cores and calculated relative permeability data are used to construct input and output datasets. The training and testing sets are set at an 8:2 ratio. Through model training and parameter adjustment, a mapping from 3D image features to relative permeability results is constructed, achieving rapid presentation of relative 3D rock core images to relative permeability. The architecture diagram of the algorithm implementation is shown in Fig. 3.

To accurately characterize the pore interface characteristics of porous media and the influence of image features on relative permeability, data such as contact angle range, interface tension, and CT scanning accuracy were embedded into a deep network structure. Through this deep embedding mechanism, the model was forced to focus on these important physical parameters. In addition, the flow equation, phase permeability law, and expert experience are integrated into the loss function of the neural network. The physical loss term is combined with the data loss term, and the loss function of implicit physical laws is used to guide the training of the neural network and achieve the embedding of physical information. The algorithm architecture diagram is shown in Fig. 4.

The embedding method based on the loss function is achieved by adding physical information-related terms to the loss function of the

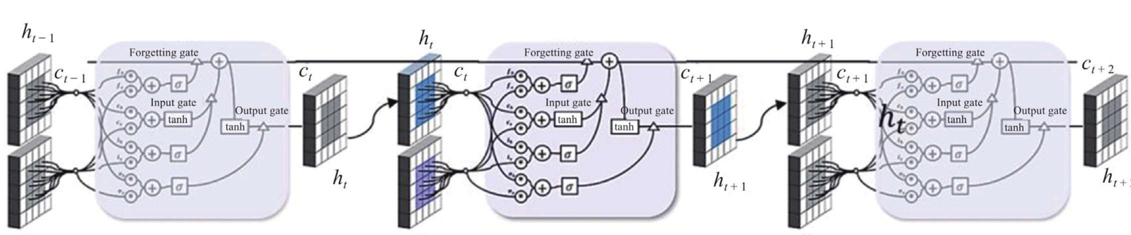

Fig. 1. ConvLSTM model architecture diagram.

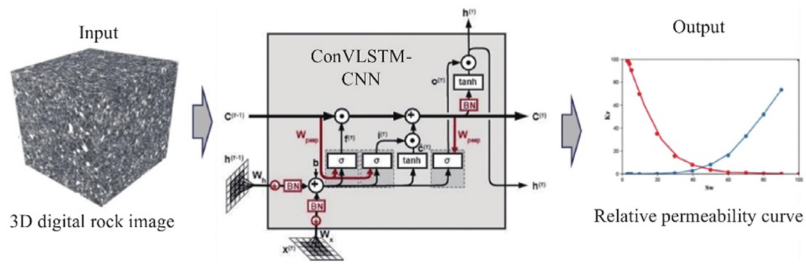

Fig. 3. Architecture diagram of the algorithm implementation.

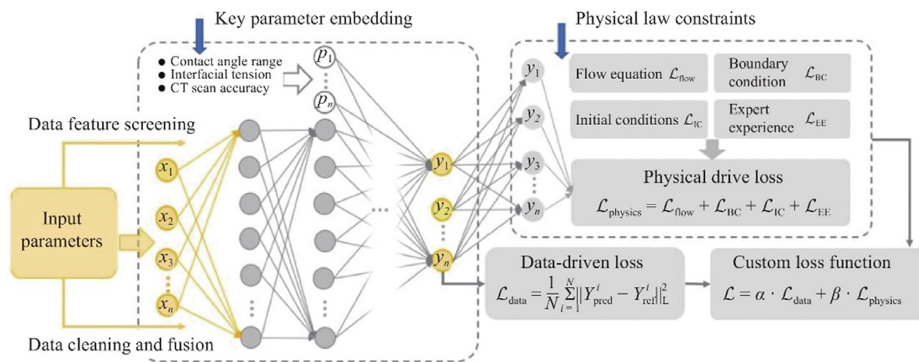

Fig. 4. Algorithm architecture diagram for embedding physical information.

model. Specifically, this means that the training of the model not only needs to minimize data errors (such as the difference between predictions and actual observations measured by root mean square error) but also needs to minimize physical errors (i.e., the degree to which the model's predictions do not match the physical equations). Physical errors can be added to the loss function in the following ways.

The loss term based on the flow equation and physical laws can be expressed as follows:

$$\mathcal{L}_{\text{flow}} = \frac{1}{N}\sum_{i=1}^{N}\left|\left|f_{\text{flow}}\left(x_i, p_i, t_i\right)\right|\right|^2_L, \tag{2.7}$$

where N is the sample size and $f_{\text{flow}}(\cdot)$ represents the residual of the flow equation or other physical laws, which may include but are not limited to equations such as fluid dynamics, heat transfer, or conservation of mass. It is a function that takes position x_i, physical parameters p_i (such as velocity, pressure, etc.), and time t_i as input and outputs a vector that represents the difference between the model's predictions and the actual physical laws given the input. $||\cdot||$ represents the Euclidean norm, which is used to calculate the residual size.

Based on boundary conditions, the loss term can be executed in the following form Eqs. (2.8):

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_{\text{BC}}}\sum_{j=1}^{N_{\text{BC}}}\left|\left|y'_{\text{BC}}\left(x_j, t_j\right) - y_{\text{BC}}\left(x_j, t_j\right)\right|\right|^2_L, \tag{2.8}$$

where N_{BC} is the number of samples on the boundary; $y'_{\text{BC}}(x_j, t_j)$ represents the predicted value of the model on the boundary; $y_{\text{BC}}(x_j, t_j)$ denotes the given values of boundary conditions; and x_j and t_j represent spatial position and time, respectively.

Based on the initial conditions, it can be executed through the following form of loss term,

$$\mathcal{L}_{\text{IC}} = \frac{1}{N_{\text{IC}}}\sum_{j=1}^{N_{\text{IC}}}\left|\left|y'_{\text{IC}}\left(x_j, t_0\right) - y_{\text{IC}}\left(x_j, t_0\right)\right|\right|^2_L, \tag{2.9}$$

where N_{IC} is the number of samples at the initial moment; $y'_{\text{IC}}(x_j, t_0)$ is the predicted value of the model at the initial stage; $y_{\text{IC}}(x_j, t_0)$ is a reference value based on initial conditions; and x_j and t_0 represents the spatial position and initial time, respectively.

Based on prior knowledge or expert experience, the following forms of loss terms can be used for execution,

$$\mathcal{L}_{\text{EE}} = \frac{1}{N_{\text{EE}}}\sum_{j=1}^{N_{\text{EE}}}\left|\left|y'_{\text{EE}}\left(x_j, t_j\right) - y_{\text{EE}}\left(x_j, t_j\right)\right|\right|^2_L, \tag{2.10}$$

where N_{EE} is the sample size based on expert experience; $y'_{\text{EE}}(x_j, t_j)$ is the predicted value of the model-related parameters; $y_{\text{EE}}(x_j, t_j)$ is a given value based on expert experience; and x_j and t_j represent the spatial position and time, respectively.

The physical driving loss function $\mathcal{L}_{\text{physics}}$, which combines the flow equation, boundary conditions, initial conditions, and expert experience, can be expressed as:

$$\mathcal{L}_{\text{physics}} = \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{IC}} + \mathcal{L}_{\text{EE}}. \tag{2.11}$$

In addition to the physically driven loss term, the data-driven loss term $\mathcal{L}_{\text{data}}$ is shown as:

$$\mathcal{L}_{\text{data}} = \frac{1}{N}\sum_{i=1}^{N}\left|\left|Y^i_{\text{pred}} - Y^i_{\text{ref}}\right|\right|^2_L, \tag{2.12}$$

where Y^i_{pred} represents the predicted output parameters of the model and Y^i_{ref} is the actual reference data.

The final loss function $\mathcal{L}$ of the model is formed by combining the data-driven loss term and the physically driven loss term,

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{data}} + \beta \cdot \mathcal{L}_{\text{physics}}, \tag{2.13}$$

where α and β are the balance coefficients of data-driven loss and physically driven loss, used to control the relative proportion of the two loss terms. The core of how the loss function can affect the operation of neural networks is that the network needs to update the weights w and biases b between each neuron based on the loss function, and the update formula generally follows gradient descent,

$$w \ := \ w - lr\frac{\partial \mathcal{L}}{\partial w} b \ := \ b - lr\frac{\partial \mathcal{L}}{\partial b}, \tag{2.14}$$

Table 2
The input parameters for the PNM simulations.

Parameters	Value
Contact angle intervals (°)	0–30, 30–50, 50–70, 70–90, 90–110, 110–130, 130–150
Interfacial tension (mN/m)	10, 20, 30
Water viscosity (cp)	1.05
Oil viscosity (cp)	2

where lr represents the learning rate of the model. The core principle of the weight and bias gradient calculation as:

$$\begin{cases} \dfrac{\partial \mathcal{L}}{\partial w_{i,j}^{(k)}} = \dfrac{\partial \mathcal{L}}{\partial Y_i^{(k)}} \dfrac{\partial Y_i^{(k)}}{\partial net_i^{(k)}} \dfrac{\partial net_i^{(k)}}{\partial w_{i,j}^{(k)}}, \\ \dfrac{\partial \mathcal{L}}{\partial b_i^{(k)}} = \dfrac{\partial \mathcal{L}}{\partial Y_i^{(k)}} \dfrac{\partial Y_i^{(k)}}{\partial net_i^{(k)}} \dfrac{\partial net_i^{(k)}}{\partial b_i^{(k)}}, \end{cases} \tag{2.15}$$

$i \in \{1, 2, \ldots, m_k\}$, $j \in \{1, 2, \ldots, m_{k-1}\}$,

where k represents the k-th layer of the neural network, m_k represents the number of neurons in the k-th layer, net represents the input vector of the neuron, and Y_i represents the output of the neuron.

The input data of all deep learning models include the 3D digital rock images, the contact angle interval, the interfacial tension, and the viscosity of oil and water, as shown in Table 2. In this paper, we set 8 groups of contact angle intervals covering from strongly water-wet to strongly oil-wet: 0° to 30°, 30° to 50°, 50° to 70°, 70° to 90°, 90° to 110°, 110° to 130°, 130° to 150°, and 150° to 180°. In addition, we choose 3 values of interfacial tension: 10 mN/m, 20 mN/m, and 30 mN/m. In each network, the contact angles are randomly distributed within the contact angle interval. Each curve consists of 10 data points. The output data are the oil-water relative permeability curves, which consist of 30 neurons divided into three sets of water saturation S_{w}, oil relative permeability K_{ro}, and water relative permeability K_{rw}. Each dataset consists of 10 neurons. They are determined by the grid search within a reasonable range, which ensures the smallest prediction error.

3. Agent construction for intelligent prediction of relative permeability

Although the ConvLSTM-CNN algorithm can achieve fast calculation of the relative permeability of rock cores, the process from digital rock cores to pore network model construction, simulation analysis result generation, and model training is relatively fragmented and requires professional personnel to implement. To achieve a more simplified and intelligent prediction of relative permeability, a set of intelligent agents for rock core relative permeability prediction is constructed using the generalization ability and task organization and scheduling ability of the LLMs, with the ConvLSTM-CNN algorithm constructed in Section 2.2 as the physical information input. The prediction and analysis of relative permeability can be achieved through simple language interaction. Therefore, to enable the large model to perform complex task analysis through simple language interaction, we have developed a workflow that allows the large model to mobilize data, algorithms, and display results according to the workflow plan.

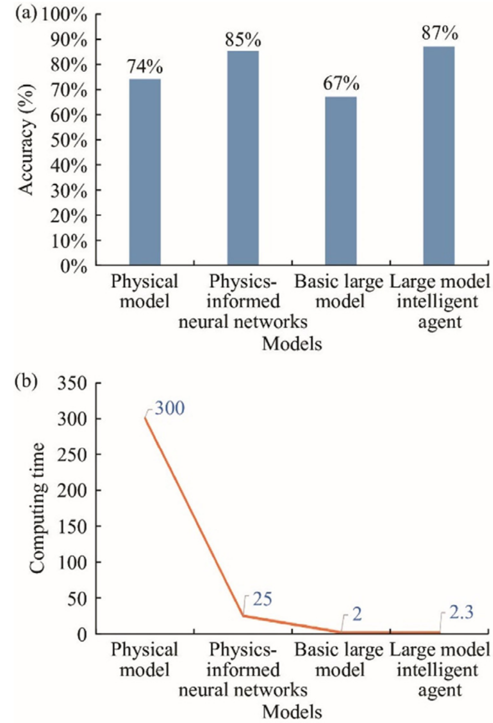

Fig. 6. Comparison of different models. (a) Comparison of model accuracy. (b) Comparison of model time consumption.

First, the first step in building an intelligent agent for predicting relative permeability is to establish a modular data and knowledge management system. On the one hand, the intelligent agent is based on the simulated data and corresponding digital core CT images constructed in the second part and preprocesses these multisource heterogeneous data, including image binary segmentation, extraction of pore topological features (such as Euler numbers and connectivity), normalization of saturation permeability curves, and noise filtering. On the other hand, by combining the physical-informed neural network (PINN) frameworks, physical constraints such as Darcy's law and conservation of mass are integrated into the network structure. The joint loss function ensures that the permeability curve output by the model is monotonic in the 0–1 interval and satisfies endpoint constraints. Meanwhile, the multitask learning strategy enables the intelligent agent to synchronously predict the permeability of the oil and water phases, improving the generalization performance of the model under different fluid conditions.

In addition, to enable the model to better match the mapping relationship between data and results, as well as optimize algorithm efficiency, during the workflow construction process, the large model

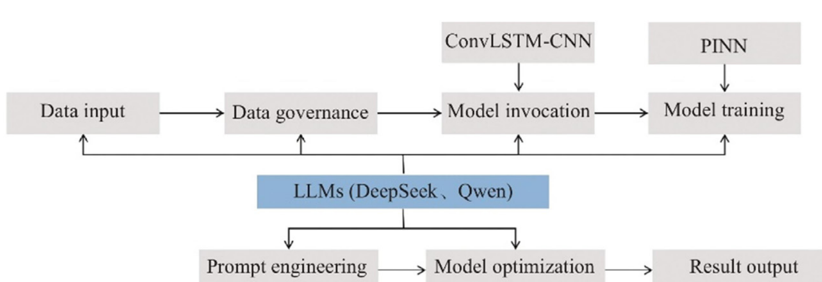

Fig. 5. Model and workflow construction process.

autonomously generates a set of prompt word engineering and then achieves the improvement of task execution through manual adjustment. The entire workflow includes data preprocessing, algorithm training, algorithm invocation, algorithm optimization and validation, and automatic generation of prompt word engineering. It can quickly adjust the workflow according to different training and analysis tasks. The model and workflow construction process are shown in Fig. 5.

4. Results and discussion

4.1. The results of the LLM-driven intelligent agent

Based on the intelligent agent constructed in Part 3 of this article, the pore network simulation is used to integrate multisource data such as three-dimensional images of rock cores, rock core properties, fluid properties, and pore structures. The ConvLSTM-CNN algorithm is used to systematically analyze the rock fluid interaction law. The process includes data preprocessing (cleaning, normalization), feature engineering (extracting key parameters such as pore distribution and wettability), model training and validation (optimizing using historical rock core experimental data), and ultimately generating high-precision relative permeability curves. The entire process is implemented through a large model, and researchers can output results through simple language interaction. The intelligent agent can also continuously optimize the prediction ability through dynamic learning, greatly simplifying the analysis process. The specific process is shown in the Appendix.

4.2. Comparison and different model results

To verify the effectiveness of the research model, experimental data from the same core were compared with traditional numerical simulations, the deep learning model constructed in Part 2, the intelligent model driven by the large model constructed in Part 3, and the basic large model. The results showed that compared with traditional numerical simulation under the same calculation conditions, both the ConvLSTM-CNN algorithm constructed in Part 2 and the intelligent agent constructed in Part 3 improved the accuracy of reservoir relative permeability prediction by approximately 10%. The basic LLMs had the lowest accuracy because they did not have deep learning of relevant physical information. In addition, the computational efficiency of the intelligent agent constructed in this paper is 10 times that of traditional deep learning algorithms, and the computational efficiency is 100–1000 times higher than that of traditional physical models.

In addition, we compared the analysis results of different basic large models without the assistance of a physical knowledge base, and the

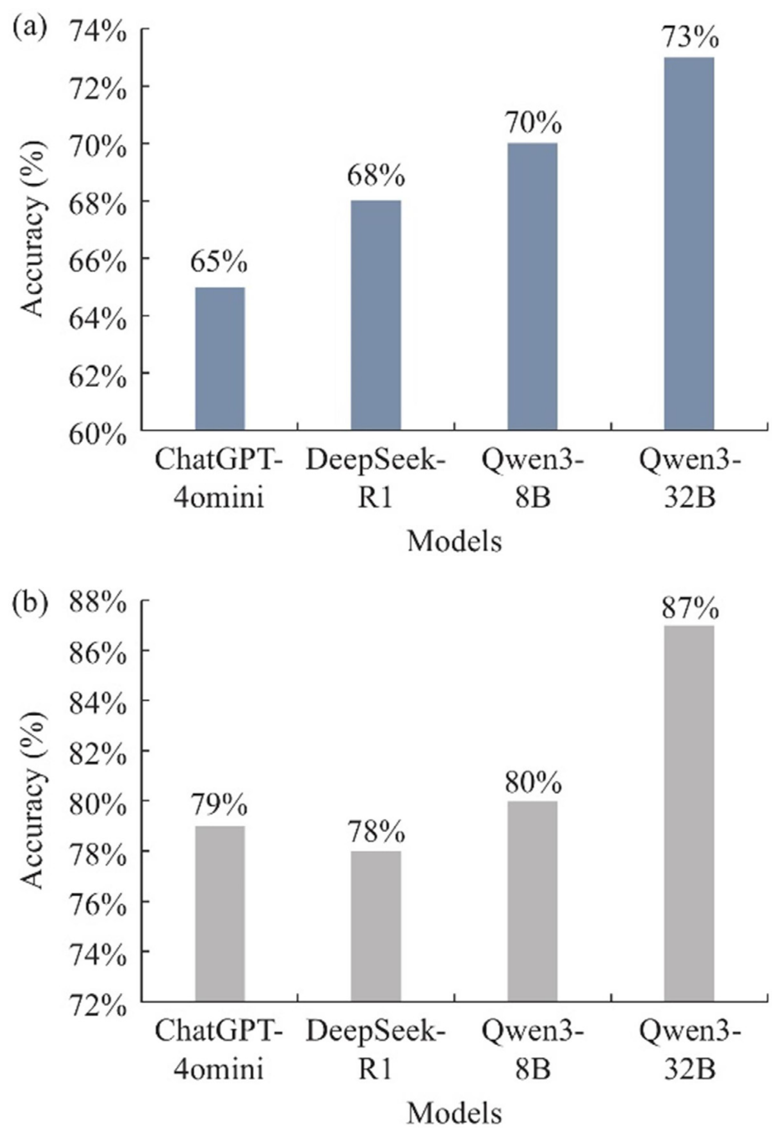

Fig. 7. Comparison of different basic larger models. (a) With model training. (b) Without model training.

results showed that the larger the model parameters were, the more accurate the results, as shown in Fig. 6(a) and (b). To further verify the adaptability of different models in this study, a comparative analysis was conducted on the analysis performance of four models with and without model training, and the results show that the DeepSeek and Qwen3 models have stronger capabilities in multimodal data analysis than ChatGPT-4omini, as shown in Fig. 7(a) and (b).

5. Conclusions

This article proposes a collaborative analysis framework for the relative permeability of oil and gas reservoirs based on intelligent agents. To address the problem of information fragmentation between macro geological modeling and micro mechanical characterization in traditional single data sources and manual analysis modes, a multiagent collaborative workflow is constructed. First, based on digital core images, a pore network model is used to obtain the relative permeability of 1000 images. Subsequently, the intelligent agent automatically completes data preprocessing, feature extraction, and parameter calibration based on preset task dependencies, ensuring consistency and comparability between macroscopic geological attributes and microscopic mechanical indicators. By introducing deep learning-based scheduling strategies, the system is able to dynamically allocate computing resources and adjust model parameters in real time, significantly improving analysis efficiency and accuracy.

This research has shown that both the ConvLSTM-CNN algorithm constructed in Part 2 and the intelligent agent constructed in Part 3 improved the accuracy of reservoir relative permeability prediction by approximately 10%. The basic LLMs had the lowest accuracy because they did not have deep learning of relevant physical information. In addition, the computational efficiency of the intelligent agent constructed in this paper is 10 times that of traditional deep learning algorithms, and the computational efficiency is 100-1000 times higher than that of traditional physical models. In addition, we compared the analysis results of different basic LLMs without the assistance of a physical knowledge base, and the results showed that the larger the model parameters were, the more accurate the results. The DeepSeek and Qwen3 models have stronger capabilities in multimodal data analysis than ChatGPT-4o.

This method has the following advantages: first, it has a high degree of automation, which can significantly shorten the data processing and model building cycle; second, it has strong flexibility, and intelligent agents can adaptively optimize for different reservoir types and experimental conditions; third, it has good scalability and can access more geophysical and chemical logging data in the future, further improving the reservoir characterization system. However, current research still has shortcomings, such as relying on prior knowledge for agent model training and limited on-site application verification cases.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jiulong Wang: Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Yuanchun Zhou:** Writing – review & editing, Supervision. **Xuhui Zhang:** Visualization, Methodology. **Junming Lao:** Validation, Investigation. **Hongqing Song:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (Grant No. 52274027) and the China Postdoctoral Science Foundation (Grant No. 2022M713204).

Appendix

The analysis steps: The first step is to input questions (as shown in Fig. 8(a)), and the intelligent agent determines the analysis magic table through semantic recognition, automatically retrieves the analysis model (as shown in Fig. 8(b)), quickly calculates the relative permeability results of the corresponding rock core through image input, and provides detailed explanations (as shown in Fig. 8 (c)).

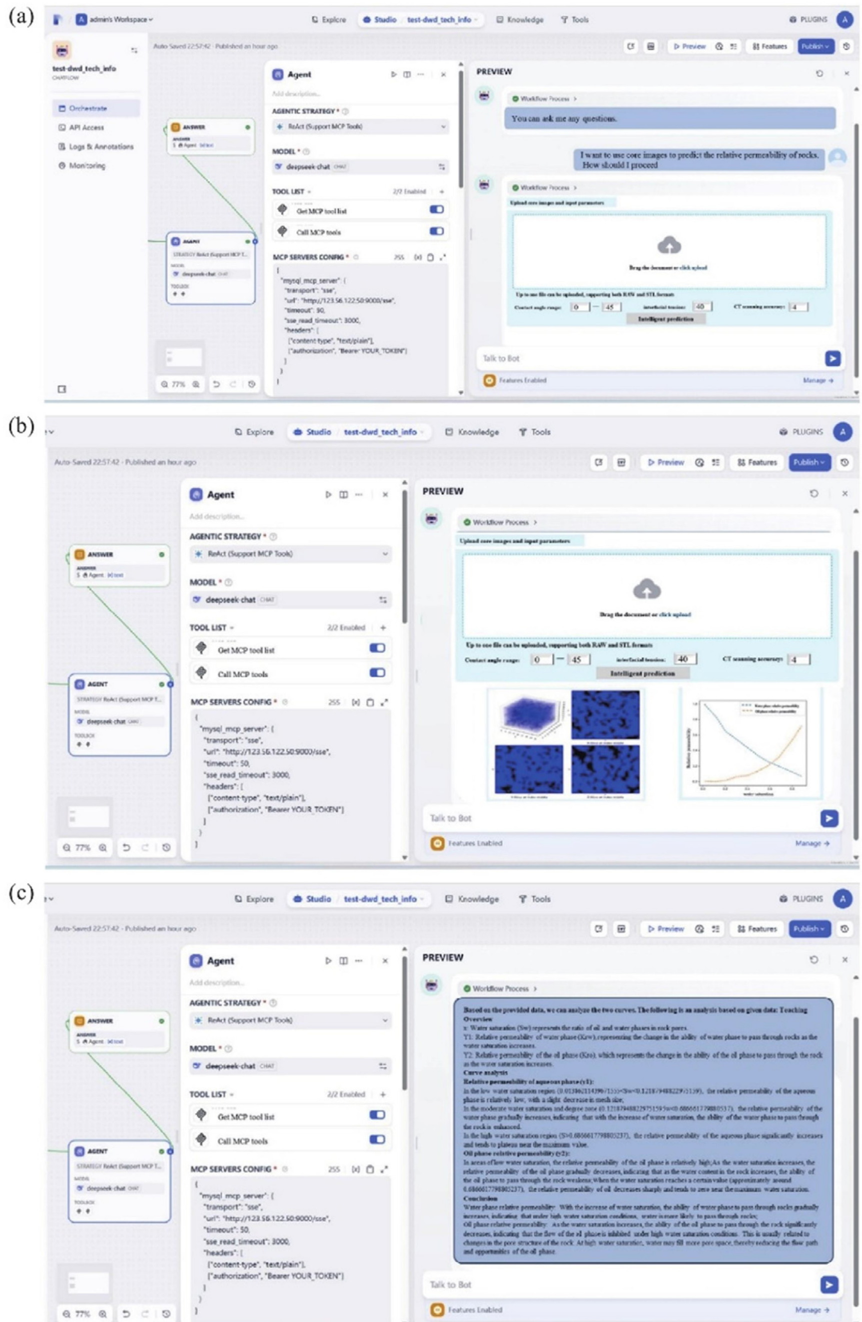

Fig. 8. Oil and gas relative permeability prediction based on the agent. (a) Step 1: Q&A. (b) Step 2: Model call. (c) Step 3: Result output.

References

[1] M. Aslannezhad, M. Sayyafzadeh, D. Tang, et al., Upscaling relative permeability and capillary pressure from digital core analysis in Otway formation: considering the order and size effects of facies, Gas Sci. Eng. 128 (2024) 205363.
[2] B. Zhao, R. Ratnakar, B. Dindoruk, et al., A hybrid approach for the prediction of relative permeability using machine learning of experimental and numerical proxy SCAL data, SPE J 25 (05) (2020) 2749–2764.
[3] S.A. Alarifi, Oil-Water Relative Permeability Prediction Using Machine Learning, SPE Middle East Oil and Gas Show and Conference (p. D011S033R008), SPE, 2023.
[4] X. Cao, B. Wu, Artificial intelligence and multiscale numerical simulation methods for intelligent oilfield reservoir modeling and their applications, Geogr. Res. Bull. 4 (2025) 229–232.
[5] J. Chung, R. Ahmad, W. Sun, et al., Prediction of effective elastic moduli of rocks using graph neural networks, Comput. Methods Appl. Mech. Eng. 421 (2024) 116780.
[6] F. Jiang, Y. Guo, T. Tsuji, et al., Upscaling permeability using multiscale X-ray-CT images with digital rock modeling and deep learning techniques, Water Resour. Res. 59 (3) (2023) e2022WR033267.
[7] T. Ma, Y. Nakamori, Modeling technological change in energy systems–From optimization to agent-based modeling, Energy 34 (7) (2009) 873–879.
[8] A. Mahdy, W. Zakaria, A. Helmi, et al., Machine learning approach for core permeability prediction from well logs in Sandstone Reservoir, Mediterranean Sea, Egypt, J. Appl. Geophys. 220 (2024) 105249.
[9] A.A. Shehata, M. Ahmed, A.A. Kassem, et al., Optimizing permeability and porosity prediction with advanced machine learning: a case study unlocking the complexities of late cretaceous reservoirs, gulf of Suez, Egypt, J. Afr. Earth Sci. (2025) 105670.
[10] Y. Sun, S. Pang, J. Zhang, et al., Porosity prediction through well logging data: a combined approach of convolutional neural network and transformer model (CNN–transformer), Phys. Fluids 36 (2) (2024) 026604.
[11] X. Xiang, Y. Chen, E. Zhang, et al., Lightweight permeability prediction of digital rocks by merging 3D depthwise separable convolution with efficient multiscale attention, Comput. Geosci. 29 (3) (2025) 1–18.
[12] P. Xu, H. Zhou, X. Liu, et al., Permeability prediction using logging data in a heterogeneous carbonate reservoir: a new self-adaptive predictor, Geoenergy Sci. Eng. 224 (2023) 211635.

[13] T. Zarin, H.A. Eshkaftaki, A. Sharifi, Machine Learning-Based Prediction of Oil–Water Relative Permeability Using Core Flooding and CT-Scan Data, Results Eng. (2025) 105735.

[14] H. Li, B. Aslam, B. Yan, An integrated deep learning and physics-constrained upscaling workflow for robust permeability prediction in digital rock physics, SPE J. (2025) 1–20.

[15] A.P. Mabiala, Z. Cai, A.K.F. Kouassi, et al., Integrating advanced machine learning models for accurate prediction of porosity and permeability in fractured and vuggy carbonate reservoirs: insights from the tarim basin, Northwest. China SPE J. (2025) 1–27.

[16] Y. Meng, J. Jiang, J. Wu, et al., Transformer-based deep learning models for predicting permeability of porous media, Adv. Water Resour. 179 (2023) 104520.

[17] A. Moradi, J. Tavakolifaradonbe, B.M. Moldestad, Data-driven proxy models for improving advanced well completion design under uncertainty, Energies 15 (20) (2022) 7484.

[18] Y. Wang, H. Li, J. Xu, et al., Machine learning assisted relative permeability upscaling for uncertainty quantification, Energy 245 (2022) 123284.

[19] Y.J. Wang, Y. Jin, B.T. Lin, et al., Deep learning-based upscaling for CO_2 injection into saline aquifers, Pet. Sci. (KeAi Commun. Co.) 4 (2025) 22.

[20] L. Yang, S. Fomel, S. Wang, et al., Porosity and permeability prediction using a transformer and periodic long short-term network, Geophysics 88 (1) (2023) WA293–WA308.

[21] Y. Zhang, Z. Li, H. Wu, Interactive machine learning for segmenting pores of sandstone in computed tomography images, Gas Sci. Eng. 126 (2024) 205343.

[22] Y. Han, F.P. Hamon, S. Jiang, et al., Surrogate model for geological CO_2 storage and its use in hierarchical MCMC history matching, Adv. Water Resour. 187 (2024) 104678.

[23] M. Elmorsy, W. El-Dakhakhni, B. Zhao, Rapid permeability upscaling of digital porous media via physics-informed neural networks, Water Resour. Res. 59 (12) (2023) e2023WR035064.

[24] R.P. Altamar, Modeling hydrocarbon pathways from accumulation to source: A stochastic agent-based approach with permeability-driven matrices, Lead. Edge 44 (7) (2025) 566–570.

[25] Z. Cao, C. Ma, W. Tang, et al., A new vision transformer model for lithofacies identification in cores, Geoenergy Sci. Eng. 240 (2024) 213012.

[26] S. Geng, S. Zhai, C. Li, Swin transformer based transfer learning model for predicting porous media permeability from 2D images, Comput. Geotech. 168 (2024) 106177.

[27] A. Kazemi, M. Esmaeili, Reservoir surrogate modeling using U-Net with vision transformer and time embedding, Processes 13 (4) (2025) 958.

[28] K. Li, H. Li, A. Khudhair, et al., Efficient generation of digital rock CT images using LoRA-enhanced stable diffusion models, Intell. Geoengin. 2 (2) (2025) 96–108.

[29] W. Liu, Y. Guan, X. Yan, et al., An efficient prediction and historical matching method for reservoir dynamics using advanced attention-based vision transformer, Phys. Fluids 37 (1) (2025) 016613.

[30] Z. Ren, S. Srinivasan, D. Crandall, Constrained Transformer-Based Porous Media Generation to Spatial Distribution of Rock Properties. arXiv preprint arXiv. (2024) 2410.21462.

[31] F. Zhang, L. Nghiem, Z. Chen, Evaluating reservoir performance using a transformer-based proxy model, Geoenergy Sci. Eng. 226 (2023) 211644.

[32] C. Xie, J. Zhu, J. Wang, et al., Direct prediction of relative permeability curve from 3D digital rock images based on deep learning approaches, Int. J. Rock Mech. Min. Sci. 170 (2023) 105544.

[33] C. Xie, J. Zhu, H. Yang, et al., Relative permeability curve prediction from digital rocks with variable sizes using deep learning, Phys. Fluids 35 (9) (2023) 096605.